



Chemical composition of acid–base fractions separated from biooil derived by fast pyrolysis of chicken manure

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ARTICLE INFO

Article history:

Received 21 March 2008

Received in revised form 17 November 2008

Accepted 25 June 2009

Available online 30 July 2009

Keywords:

Biooil

Fatty acids

Heterocyclic-N

Sterols

Long-chain aliphatics

ABSTRACT

Our earlier investigations on the chemical composition of biooils derived by the fast pyrolysis of chicken manure revealed the presence of more than 500 compounds. In order to simplify this heterogeneous and complex chemical system, we produced four biooil fractions namely strongly acidic fraction A, weakly acidic fraction B, basic fraction C and neutral fraction D on the basis of their solubilities in aqueous solutions at different pHs. The yield (wt/wt.%) for fraction A was 3%, for fraction B 21.3%, for fraction C 2.4% and for fraction D 32.4%, respectively. The four fractions were analyzed by elemental analyses, Fourier Transform infrared spectrophotometry (FTIR), ¹H and ¹³C nuclear magnetic spectroscopy (NMR), and electrospray ionization mass spectrometry (ESI-MS). The major components of the four fractions were saturated and unsaturated fatty acids, N-heterocyclics, phenols, sterols, diols and alkylbenzenes. The pH separation system produced fractions of enhanced chemical homogeneity.

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1. Introduction

Biooil (also known as bioliquid) production has recently gained importance because of the need to reduce environmental contamination and dependency of fossil fuels, and for its potential to generate economic value for waste residues. Biooils from any waste biomass such as forestry biomass, crop residues or animal manures contain low amounts of sulfur, are always carbon neutral and, most importantly, unlike petroleum feedstock, are renewable. With an increasing population, more and more biomass residues are being generated, due to higher demands for food and shelter. In North America, the production of some animals has increased due to the higher demand for protein in international markets (Statistics Canada, 2006). Increased animal production is paralleled by higher risks of water and air contamination with chemical and biological agents and greater emissions of greenhouse gases (El-Ahraf and Willis, 1996; Lusk et al., 1997; Miner et al., 2000). Development of advanced state-of-the-art commercial technologies such as fast pyrolysis will help treating the waste generated thereby decreasing

environmental pollution, and permitting conversion of agricultural biomass into useful bioproducts.

The chemical composition of biooils is determined by the nature of the biomass from which they originate. For example, biooils obtained from forestry biomass are rich in oxygen containing compounds, which polymerize with time to make the biooils unstable (Czernik et al., 1994; Diebold and Czernik, 1997). Our recent research has focused on the chemical characterization of biooils derived by fast pyrolysis of chicken manure. Two biooil fractions (an aqueous and a viscous) were characterized by FTIR, NMR and three mass spectrometric methods (Schnitzer et al., 2007a,b, 2008). More than 500 different compounds were identified by pyrolysis field ionization- and field desorption mass spectrometry (Py-FIMS and Py-FDMS) in both biooil fractions. For example, the viscous fraction contained relatively higher amounts of N-heterocyclics than the aqueous fraction (Schnitzer et al., 2007b). Some of the N-heterocyclics identified by the different mass spectrometric methods are high-value feedstocks (i.e., pyrazines and pyrroles) frequently used as feedstocks in the chemical and pharmaceutical industries. The latter prompted us to develop methods for extracting and separating chemicals from complex biooils. In this article, we report on the separation and chemical characterization of groups of chemicals from a viscous biooil fraction based on their solubility in aqueous solutions of different pH values. Our main objective was to simplify the complex chemical biooil system so that we could proceed to the isolation of organic components of commercial interest.

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2. Experimental

2.1. Production of biooils from broiler chicken manure

Manure samples were collected from a broiler chicken farm located 30 km South of Ottawa, Canada. Biooil Fractions I (aqueous fraction) and II (viscous fraction) were produced by Advanced Bio Refinery Inc., Ottawa, Canada using its proprietary fast pyrolysis technology. The commercial 1 ton/day pyrolyzer equipment and the detailed pyrolysis process have been described elsewhere (Schnitzer et al., 2007a). The biooil fractions used for the current study were, however, produced using a laboratory scale (approximately 1 kg of dried broiler chicken manure for each run) bench-top set up operating under the same principle as described by Schnitzer et al. (2007a). Broiler chicken manure used for biooil production was dried at 100 °C overnight, ground to <1 mm size particles, transferred to a bench top pyrolyzer preheated at 350 °C and exposed to this temperature for 1–2 s. The vapors produced by the rapid thermal conversion of chicken manure were cooled inside a condenser. The resulting biooil produced two distinct layers when allowed to stand for a day at room temperature, a viscous bottom layer and an aqueous top layer. The two fractions were separated by decantation and stored at room temperature for further analysis. Most data reported in this paper were obtained on the viscous biooil fraction which will be referred to as biooil.

2.2. Chemical analysis

The carbon, hydrogen, nitrogen and sulfur contents of Fraction II biooil and the four fractions derived from it were determined by dry-combustion on a LECO CHNS-932 CHNS analyzer. All results for the elemental analyses were expressed on a moisture and ash-free basis. Phosphorous content of the viscous biooil was analyzed by inductively coupled plasma mass spectrometry. Oxygen was calculated by subtracting the percentages of C, H, N, S and P from 100%. All samples were analyzed in duplicates with standard devi-

ations of less than $\pm 5\%$. Moisture was determined by heating samples at 105 °C for 25 h, and ash by heating moisture-free samples at 700 °C for 6 h.

2.3. FTIR analysis

FTIR spectra of the biooil and pH-separated fractions were recorded on a Perkin Elmer Spectra BX II spectrophotometer. The samples were placed between two ZnSe windows of 3 mm thickness and scanned from 400 to 4000 cm^{-1} at a spectral resolution of 2 cm^{-1} . The spectra were recorded as averages of 16 consecutive scans. The interpretations of the FTIR spectra were based on data presented by Silverstein and Webster (1998), Bellamy et al. (1956) and Nakanishi and Solomon (1977).

2.4. NMR analysis

^{13}C and ^1H NMR analyses of the four fractions separated from the biooil were recorded on a Bruker AMX instrument operating at 400 MHz. In a typical sample preparation procedure, the samples were freeze-dried for 24 h and the resulting solids dissolved in deuterated methanol at a concentration of more than 0.5 g/mL. Chemical shifts were referenced to tetramethylsilane for both ^{13}C and ^1H NMR spectra. The chemical shifts in ^1H NMR spectra of the Fractions A, B, C and D were classified as follows (Table 3): 0–2.9 ppm (aliphatic protons), 3.0–4.1 ppm (carbohydrate protons), 4.2–6.0 ppm (phenolic protons), 6.0–8.0 ppm (aromatic and/or N-heterocyclic protons), 8–10 ppm (N-heterocyclic, –CHO and/or –COOH protons) (Schnitzer et al., 2007a; Marche et al., 2005; Ingram et al., 2008). Likewise, the chemical shifts in ^{13}C NMR spectra of the four fractions were divided into the following regions: 10–40 ppm (paraffinic C), 41–60 ppm (C in OCH_3 , amino acids and sterols), 61–105 ppm (C in carbohydrates and aliphatics with OH groups), 106–150 ppm (aromatic C, C in aromatic N-heterocyclics, and sterols), 151–170 ppm (phenolic C) and 171–190 ppm (C in COOH, amide and ester groups) (Wehrli and Wirthlin, 1978).

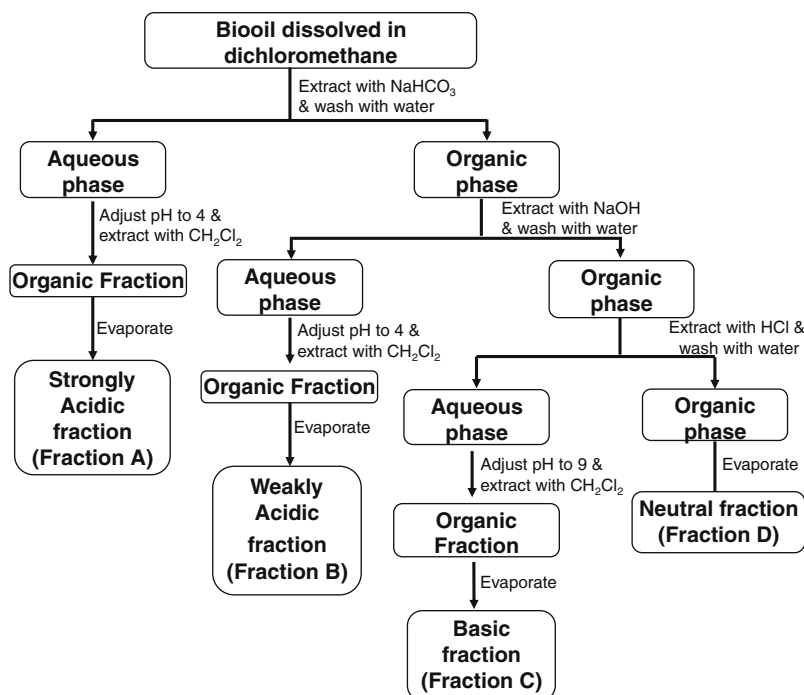


Fig. 1. Acid–base separation of the viscous biooil.

2.5. Separation of Fractions A to D by acid-base extraction

The procedure for the acid–base extraction of chemical components from the biooil is summarized in Fig. 1. This procedure is a slight modification of the method used by Dote et al. (1992) for the chemical fractionation of oil from sewage sludge. Briefly, aqueous solutions of sodium bicarbonate, sodium hydroxide and hydrochloric acid were successively used to extract acidic, basic and neutral components from a dichloromethane solution of the biooil. The components dissolved in the aqueous solutions were then re-extracted into dichloromethane and denoted as strongly acidic (Fraction A), weakly acidic (Fraction B), basic (Fraction C) and neutral (Fraction D).

The experimental details were as follows: 20 g of the biooil was dissolved in 500 mL of dichloromethane. Any insoluble char in Fraction II was removed by filtration through a G2 glass fiber filter (1 µm Fisherbrand) paper using a Buchner filtration system. The filtrate was taken to dryness using a Buchi rotavapor (R-200) fitted with a Buchi distillation chiller (B-741) to yield a char-free biooil. Approximately, 10 g of the latter was then re-dissolved in 500 mL of dichloromethane and transferred to a separating funnel. The organic compounds in the solution were first extracted with 200 mL of 5% aqueous sodium bicarbonate solution by manual shaking for about 5 min. The procedure was repeated twice more with 200 and 100 mL of 5% aqueous sodium bicarbonate solution (total volume 500 mL). The aqueous and the organic layers were allowed to settle and then separated. The organic layer was further washed with 100 mL (50 mL × 2) deionized water (shaking of ~2 min) and the washings were combined with the aqueous extract. The latter was acidified to pH 4 with hydrochloric acid and re-extracted into dichloromethane, which was dried to yield strongly acidic Fraction A (Fig. 1). The resulting organic layer was extracted with 500 mL of 5% aqueous sodium hydroxide solution following the procedure shown in Fig. 1, which produced an aqueous and an organic phase. The aqueous phase was then acidified to pH 4 and re-extracted with dichloromethane. The material obtained after drying of the organic layer was denoted as weakly acidic Fraction B. The organic phase produced from the above separation process was extracted with 500 mL of 5% hydrochloric acid following the same procedure, which resulted in an aqueous and an organic phase. Separation of aqueous layer, adjusting its pH to 9, subsequent re-extraction into dichloromethane and drying yielded basic Fraction C. The material soluble in the organic phase after drying produced the neutral Fraction D (Fig. 1).

2.6. Procedure for electrospray ionization mass spectrometric (ESI-MS) analysis

Molecular ions in solution can be analyzed by electrospray ionization mass spectrometry (ESI-MS). The detailed technical aspects of ESI-MS were reported earlier (Porter, M.Sc. Thesis, 2003). All spectra were recorded on a Micromass Quattro-LC triple-quadrupole mass spectrometer fitted with Z-spray ion source and the data were analyzed by a MassLynx data acquisition and processing soft-

ware. Typically, 20 µL of sample solution (1 mg biooil per 5 mL HPLC grade methanol) was directly introduced into the mass spectrometer with the mobile phase (1:1 v/v mixture of HPLC grade acetonitrile and methanol), running at a flow rate of 40 µL/min. The mobile phase flow rate was maintained with a HPLC flow controller. Nitrogen was used as the desolvation gas at a flow rate of 100 L/h. The inlet capillary voltage was held between +4.0 and +4.5 keV in positive ion mode, whereas in negative ion mode it was between –3.5 and –4.0 keV. The cone and the extractor voltages were adjusted as required to obtain a stable signal. ESI-MS spectra were obtained in the range of 20–1000 mass to charge ratios (m/z).

All four biooil fractions were analyzed by ESI-MS in both positive and negative modes. Compounds such as N-heterocyclics, which can accept protons to produce molecular ions can be analyzed in the positive mode; while compounds such as fatty acids release protons to form negative molecular ions which can then be determined in the negative mode. In the absence of an added metal ion salt or a polyatomic organic acid or base, one atomic mass unit (AMU) should be subtracted from m/z values of the ions obtained in the positive mode to calculate the actual mass of the corresponding compound. Similarly, one AMU should be added to the m/z of the ions obtained in the negative mode to calculate the m/z of the corresponding compound. The compounds corresponding to the m/z so obtained were identified by comparing their ESI-MS spectra with those of known compounds (Schulten and Simmleit, 1986; Schnitzer and Schulten, 1992; Marche et al., 2005; Schnitzer et al., 2007b, 2008) and with the Wiley mass spectral library software edition 6.

3. Results and discussion

3.1. Pyrolysis of chicken manure, yield of co-products and chemical characteristics of chicken manure, biooil and four separated fractions

The pyrolysis of chicken manure produced two biooil fractions, one bio-char and non-condensable gases. Some characteristics of the different products obtained from the pyrolysis of dried chicken manure are presented in Tables 1 and 2. Table 1 shows that 15 wt.% of the broiler chicken manure was converted into the aqueous biooil, whereas the viscous biooil fraction accounted for ~35 wt.%. The residual 50 wt.% of the material was converted into bio-char (27 wt.%) and non-condensable gases (23 wt.%). While some bio-char was carried over along with the vapors during the pyrolysis process and appeared in the biooil, it was removed from the biooil samples prior to the analyses. Whereas the ash content of the broiler chicken manure was 27 wt.% that of the two biooils was substantially lower (0.5 and 0.4 wt.%, respectively). The ash represented 48.6% of the total char weight (Table 1). Table 2 shows the yield and elemental composition of the initial broiler chicken manure, viscous biooil, and of Fractions A, B, C, and D obtained from the viscous biooil fraction. Yields of the various fractions were as follows: 3.0% for strongly acidic Fraction A; 21.3% for weakly acidic Fraction B; 2.4% for basic Fraction C; and 32.4% for neutral

Table 1
Yield and properties of various co-products derived by fast pyrolysis of broiler chicken manure.

Sample	Color	Phase	Yield ^a (% wt/wt)	Moisture (% wt/wt)	Ash ^b (% wt/wt)
Chicken manure	Brown	Solid	100	11.4	27.0
Biooil aqueous	Dark brown	Liquid	15	32.0	0.5
Biooil viscous	Black	Semi liquid	35	3.0	0.4
Char	Black	Solid	27	ND	48.6
Non-condensable gas	–	Gas	23	ND	–

^a Yield expressed as percentage of dry weight of chicken manure.

^b On moisture-free basis.

Table 2

Yield and elemental analyses of broiler chicken manure, viscous biooil fraction and the four fractions separated from it by acid–base separation.

Sample name	Yield (% wt/wt)	Element content (wt/wt.%)						H/C ^a	N/C ^a
		C	H	N	S	P	O ^e		
Chicken Manure	100	51.7 (0.5) ^b	5.8 (0.07)	5.9 (0.02)	0.8 (0.01)	ND ^d	35.9	1.35	0.10
Biooil viscous	35 ^c	62.4 (0.6)	7.5 (0.06)	6.3 (0.08)	0.5 (0.02)	0.04 (0.02)	23.3	1.50	0.10
Fraction A	3	64.7 (0.7)	8.1 (0.05)	10.3 (0.06)	0.4 (0.03)	ND	16.5	1.50	0.14
Fraction B	21.3	74.1 (0.7)	9.4 (0.05)	3.1 (0.01)	0.3 (0.01)	ND	13.2	1.50	0.03
Fraction C	2.4	65.5 (0.3)	8.3 (0.07)	9.0 (0.02)	0.6 (0.01)	ND	16.6	1.51	0.12
Fraction D	32.4	76.9 (0.8)	11.0 (0.09)	4.4 (0.02)	0.4 (0.03)	ND	7.3	1.72	0.04

^a Atomic ratios.^b Number in parenthesis represents standard deviation of mean.^c Biooil viscous fraction accounts for 35% of the weight of the initial chicken manure.^d Not determined.^e O Determined by difference.**Table 3**

Most prominent bands in FTIR spectra of the four fractions separated from the biooil viscous fraction.

Fractions	Wavenumbers in IR spectra (cm ⁻¹)
Fraction A	3291b ^a , 2970, 2358, 1700–1600(b), 1454, 1376, 1200–1180, 1050, 967, 879, 754, 665
Fraction B	3282(b), 3005, 2927, 2853, 1698, 1651, 1595, 1514, 1463, 1454, 1376, 1270, 1239, 1170, 1166, 1120, 1083, 989, 967, 879, 818, 754, 693
Fraction C	3241(b), 2966, 2931, 2873, 2361, 1667(b), 1505, 1455, 1380, 1267, 1234, 1200–1160, 1166, 1089, 1050, 881, 754, 668
Fraction D	3286(b), 2924, 2855, 2361, 2342, 1700–1600(b), 1462, 1378, 1169, 1050, 967, 879, 744, 668

^a b: Broad peak.

Fraction D. The total yield was approximately 60 wt.% while 40 wt.% remained in the aqueous phases after extractions.

Elemental analyses of fractions A and C showed similar carbon (~65%) and hydrogen (~8.0%) contents as that of the parent viscous biooil (called biooil herein). Fractions B and D, which were obtained in relatively high yields, showed higher carbon (~75%) and hydrogen concentrations (>9.4%) than Fractions A and C, and biooil. The higher hydrogen concentration of Fractions B and especially D indicates the presence of paraffinic aliphatics. Fractions A and C had higher N content (>9%) than fractions B and D (3.1 and 4.4, respectively). The high nitrogen concentration of Fractions A and C appears to be due to the presence of large amounts of N-heterocyclics and/or other N-compounds as reported earlier by Schnitzer et al. (2007b, 2008). The sulfur concentration of biooil and the four fractions separated from it was low (<1%), whereas the oxygen concentration ranged from 7% to 23%, and decreased in the following order: biooil > Fraction A ≈ Fraction C > Fraction B > Fraction D. The low oxygen content in Fractions D (7%) and B (13%) complements the hydrogen content data for the same fractions indicating paraffinic aliphatics. The phosphorous concentration of biooil was 0.04%. All fractions except D, showed similar H/C ratios. High N/C ratios for Fraction A and C were likely associated with the presence of relatively large amounts of N-heterocyclics. Fractions B and D showed very low N/C ratios indicating the presence of fewer N-containing components (Table 2).

3.2. FTIR Spectra of the four fractions

The FTIR spectra of the Fractions A, B, C and D are presented in Fig. 2 while major peaks of interest are shown in Table 3. Large and broad bands due to hydrogen bonded OH groups, including water were observed in all spectra between 3291 and 3241 cm⁻¹. The band narrowed somewhat from Fraction A to Fraction D. Intensities of bands between 2970 and 2924 cm⁻¹ and between 2870 and 2855 cm⁻¹, due to –CH₃ and –CH₂ groups respectively, increased and sharpened from B to D, indicating a relatively high

content of long-chain aliphatics especially in Fraction D and B. These data complement the higher hydrogen and lower oxygen content measured in Fraction D relative to the other fractions.

The small shoulder at 3005 cm⁻¹ in the spectrum of Fraction B and even weaker shoulders at the same wavelength in the other three spectra appeared to arise from CH absorptions. Broad bands between 1700 and 1600 cm⁻¹ appeared in the spectra of all fractions except Fraction B & D; the latter two showed four bands in this region. The absorbance at 1695 cm⁻¹, due to C=O of –COOH groups and that at 1645–1651 cm⁻¹ was attributed to C=O stretching vibrations of amides (Silverstein and Webster, 1998). The peak at 1603 cm⁻¹ appeared to be due to COO⁻ (Silverstein and Webster, 1998). The spectrum of Fraction B showed an additional sharp band at 1514 cm⁻¹, which could be due either to secondary aromatic amines or to C=C stretching in pyridine rings (Silverstein and Webster, 1998; Bellamy et al., 1956). Other well defined bands in all spectra were at 1463–1455 cm⁻¹ (CH₂ deformation) and 1380–1378 cm⁻¹ (CH₂ and CH₃ bending), (Silverstein and Webster, 1998). The spectrum of Fraction B showed additional peaks at 1270 cm⁻¹ (C–N stretching of aromatic N-heterocyclic compounds), and 1239 cm⁻¹ and a shoulder at 1205 cm⁻¹, both due to C–H deformation of pyridines (Nakanishi and Solomon, 1977). Bands were observed in the 1200–1160 cm⁻¹ regions which include absorbance from OH in phenols (Bellamy et al., 1956). The absorbance at 1154 cm⁻¹ in the spectra of Fraction B and D were due to ring vibrations of pyridines. All Fractions showed absorbance at 1046–1050 cm⁻¹, with Fractions A and C showing signals (C–O stretching of carbohydrates). Fractions A and B showed absorbance at 967 cm⁻¹ (ring vibrations and C–H deformation of pyrimidines). All Fractions showed bands at 879–881 cm⁻¹ (γ-CH of furans), 754 cm⁻¹ (β-rings of pyridines) and 665–668 cm⁻¹ (ring vibrations and C–H deformations of pyridines) (Nakanishi and Solomon, 1977).

The FTIR spectra of the four fractions were dominated by aliphatics and aromatic N-heterocyclics. Fractions D and B were especially high in aliphatics, i.e., alkanes, fatty acids, and alkyl substituted aromatic rings. Most of the N-heterocyclics were concentrated in Fraction C and A, although small amounts were also present in the other fractions. The FTIR spectra show that biooil is a complex mixture of many organic compounds.

3.3. ¹H NMR spectra of the four fractions

Fig. 3 shows the ¹H NMR spectra of Fractions A to D. The chemical shifts (δ) shown in each of the four spectra are summarized in Table 4. Aliphatic protons (CH₃, CH₂ and CH protons bonded to C) were prominently observed between 0.7–2.0 and 2.3–2.9 ppm in all four spectra. The chemical shifts of protons under different electronic environments appeared as follows: cyclopropyl CH–

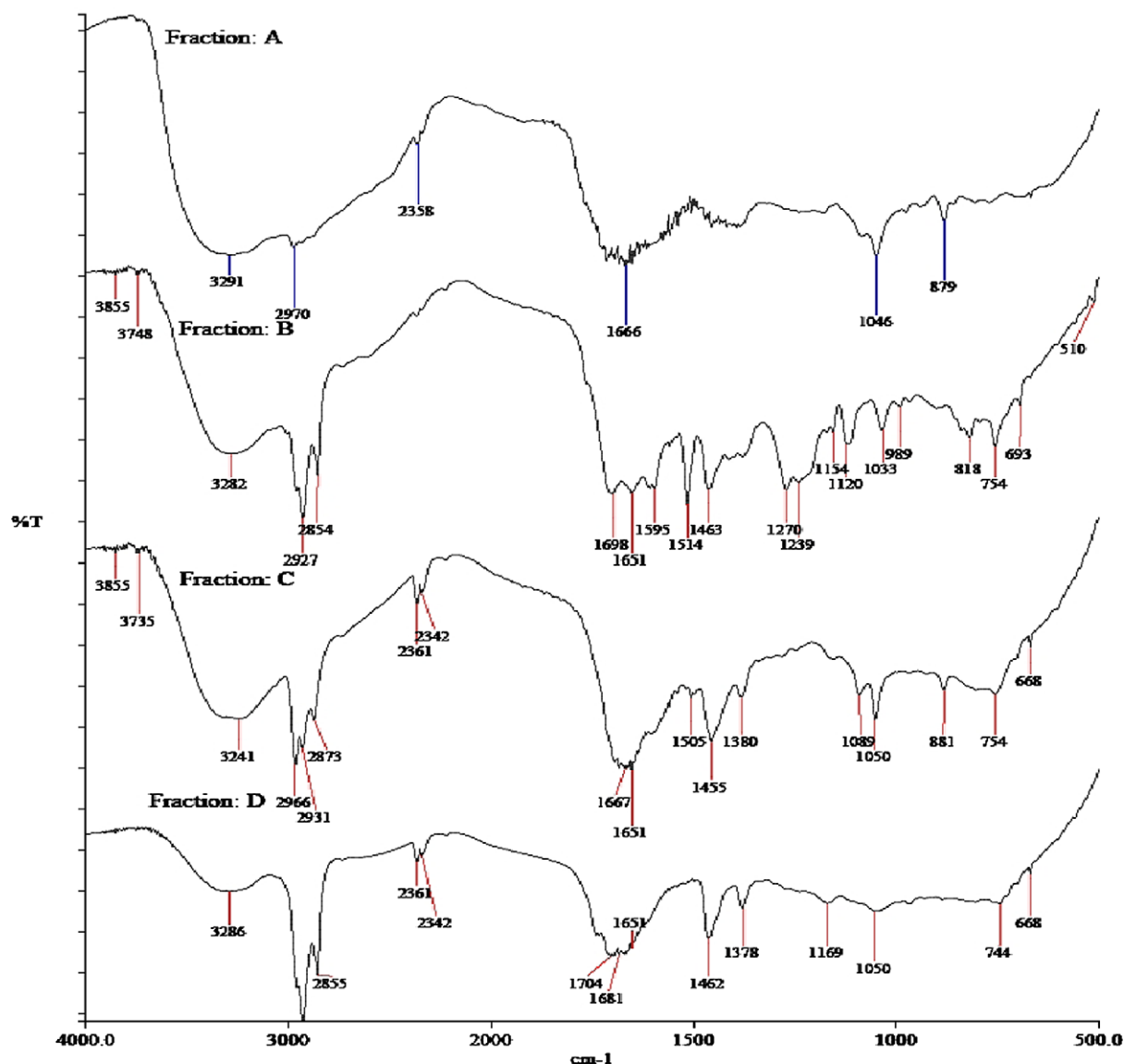


Fig. 2. FTIR spectra of fractions A to D.

(0.7 ppm), $\text{CH}_3\text{-C}$ (0.9 ppm), $\text{CH}_3\text{-C=C}$ (1.1 ppm), $\text{-C-CH}_2\text{-C}$ (1.3 ppm), $\text{CH}_3\text{-C-O}$ (1.4 ppm), $\text{CH}_3\text{-C=O}$ (1.6 ppm), $\text{-C=C(CH}_3\text{)-CO}$ (1.8 ppm) and $\text{-C-CH}_2\text{-C-O}$ (1.9 ppm) (Schnitzer et al., 2007a; Ingram et al., 2008; Nakanishi and Solomon, 1977). Of special interest are the chemical shifts at 2.3, 2.4 and 2.6 ppm, which have been assigned to $\text{CH}_3\text{-Ar}$ (Ar: aryl group), $\text{CH}_3\text{-CO-O-Ar}$ and $\text{CH}_3\text{-CO-Ar}$, respectively (Marche et al., 2005; Ingram et al., 2008; Nakanishi and Solomon, 1977). The aryl groups can be substituted by N-heterocyclic aromatics, which yield similar chemical shifts (Schnitzer et al., 2007a; Nakanishi and Solomon, 1977). The protons between 3.0 and 4.1 ppm may be due to carbohydrate residues or aliphatic compounds containing OH groups except for the protons at 3.3 ppm, which were due to deuterated methanol (CH_3OD) used as solvent to record the ^1H NMR spectra. Fraction C shows two strong proton signals at $\sim 3.8\text{--}3.9$, suggesting a high content of carbohydrate protons in this fraction. Fraction A showed signals for carbohydrate protons but to a lesser extent. All fractions showed chemical shifts between 4.8 and 5 ppm characteristic of protons in water. Although all fractions were freeze-dried before recording ^1H NMR spectra, the chemical shift for water may be due to water in the solvent (deuterated methanol) or to water of

hydration in some of the compounds in the mixture. Fraction C was the only one to show chemical shifts higher than 6.0 ppm, which were most likely due to aromatic protons in benzenoid structures and/or in aromatic N-heterocyclics. In view of the relatively high nitrogen content of Fraction C (Table 2), it is likely that the signals above 6 ppm arose from protons in N-heterocyclics such as those shown at 6.6, 6.7, 6.9, 7.0, 7.2, and 7.3 ppm in the ^1H NMR spectrum of Fraction C (Fig. 3 and Table 4). It is noteworthy that chemical shifts for N-heterocyclics have been reported to occur at 6.2, 6.7 and 8.0 ppm for indoles, and at 7.0 and 8.5 ppm for pyridines and pyrazines (Marche et al., 2005; Schnitzer et al., 2007a). Fraction A and D showed three small signals at 6.6, 7.3 and 8.4 ppm indicating the presence of small amounts of aromatic N-heterocyclics and/or aromatics in this fraction. Fraction B, with low N-content (Table 2) showed only very small proton signals above 6 ppm indicating the presence of only low concentrations of aromatic/N-heterocyclic protons.

Results obtained from the ^1H NMR analyses are consistent with those obtained from FTIR and elemental analyses, and appear indicate the following: long-chain aliphatics appear preferentially in Fraction D with minor amounts in Fraction B. Heterocyclic-N and

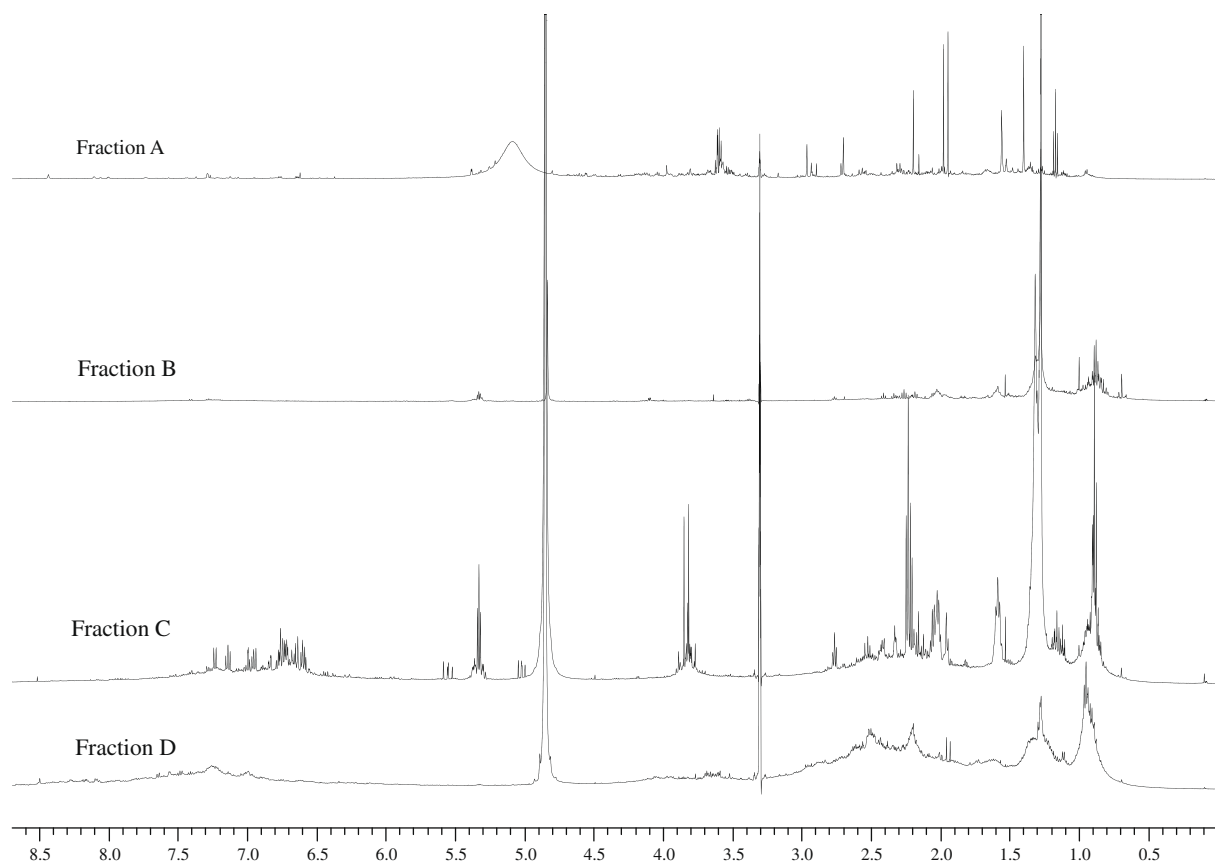


Fig. 3. ^1H NMR spectra of Fractions A to D.

Table 4

Most prominent chemical shifts (δ) in ^1H NMR spectra of the four biooil fractions.

Fraction	Chemical shifts (δ) in ppm
A	<u>1.2</u> ^a , <u>1.4</u> , <u>1.6</u> , <u>2.0</u> , <u>2.2</u> , 2.3, 2.7, 2.9, 3.0, 3.3 (methanol), <u>3.6</u> , 4.0, 5.1
B	0.7, <u>0.9</u> , <u>1.0</u> , <u>1.2</u> , 1.3, 1.5, 1.6, 2.0, <u>3.3</u> (methanol), 3.6, 4.1, <u>4.8</u> (water), 5.3
C	<u>0.9</u> , 1.1, <u>1.2</u> , <u>1.3</u> , 1.5, 1.6, 2.0, 2.1, <u>2.2</u> , 2.3, 2.4, 2.5, 2.7, 3.3 (methanol), 3.7, 3.8, <u>4.8</u> (water), 5.3, 5.6, 6.6, 6.7, 6.9, 7.0, 7.2
D	<u>0.9</u> , 1.3, 1.9, 2.0, 2.2, 2.5, <u>3.3</u> (methanol), 3.7, 3.8, 4.0, <u>4.9</u> (water), 7.0, 7.3

^a Underlined chemical shifts indicate a relatively high abundance of the corresponding protons.

other N-compounds appear concentrated in Fraction C with some also showing in fraction A. Carbohydrates appear to concentrate in Fraction D with minor amounts also showing in Fraction A.

3.4. ^{13}C NMR spectra of the four fractions

Liquid-state ^{13}C NMR spectra of Fractions A to D are presented in Fig. 4. The ^{13}C NMR spectra for the aqueous and viscous biooil fractions were published earlier (Schnitzer et al., 2007a). Information provided by the ^{13}C NMR spectra showed that Fractions A to D contained aliphatic C. Close inspections of the four Fractions spectra shows that paraffinic carbons (10–40 ppm) and C in OCH_3 groups (41 and 60 ppm) were present in all spectra. Strong signals at 50 ppm, due to carbon atom in CH_3OD were observed in all four fractions. Fractions A, B and C showed resonances between 61 and 105 ppm arising from C in carbohydrate residues and aliphatics with OH groups, C in aromatic and aromatic N-heterocyclics (106–150 ppm), phenolic C (151–170 ppm), and C in carboxylic, amide and ester groups (171–190 ppm) (Ingram et al., 2008; Schnitzer et al., 2007a; Wehrli and Wirthlin, 1978). The most abun-

dant resonances in Fraction A were at 18 and 22 ppm (terminal CH_3), 26 and 29 ppm (CH_2), and 32 ppm [$(\text{CH}_2)_n$] of long-chain aliphatics and fatty acids. Other strong signals were observed at 59 ppm, followed by resonances between 62 and 105 ppm due to C in carbohydrate residues. Relative to Fraction C, Fraction A presented fewer signals associated with C in aromatic and/or heterocyclic-N structures. The ^{13}C NMR spectrum of Fraction B was dominated by two strong signals at 18 and 59 ppm that may reflect the presence of sterols. Fraction C presented intense signals between 110 and 150 ppm (aromatic and/or aromatic N-heterocyclics), signals between 150 and 165 ppm (phenols) and 165–190 ppm (carboxyl, amide and ester groups).

Results obtained from the ^{13}C and ^1H NMR analyses are consistent with those obtained from FTIR and elemental analyses, and indicate that long-chain aliphatics appear preferentially in Fraction D but also in Fraction B. Carbohydrates appear to concentrate in Fraction D with some also showing in Fraction A. Fractions A and D contained most carbohydrate residues. Fraction B contained relative high concentration of sterols. Fraction C showed the presence of aromatic-C and/or heterocyclic-N and phenols which were also found in Fraction A to some extent (Fig. 4).

3.5. Mass spectrometric analysis

The ESI-MS data for Fractions A to D are presented in Tables 5 and 6. The identified compounds were grouped into five classes such as fatty acids, aliphatic N-compounds, N-heterocyclics, phenolics and benzenes, diols and sterols. The strongly acidic Fraction A (Table 5) showed the presence of phenolic compounds such as phenol (m/z 95), vinyl phenol (m/z 121), methyl vinyl phenol (m/z 135), guaiacol (m/z 125) and methyl guaiacol (m/z 139). Other significant components were diols, ranging from $n\text{-C}_7$ (m/z 133) to $n\text{-C}_9$ (m/z

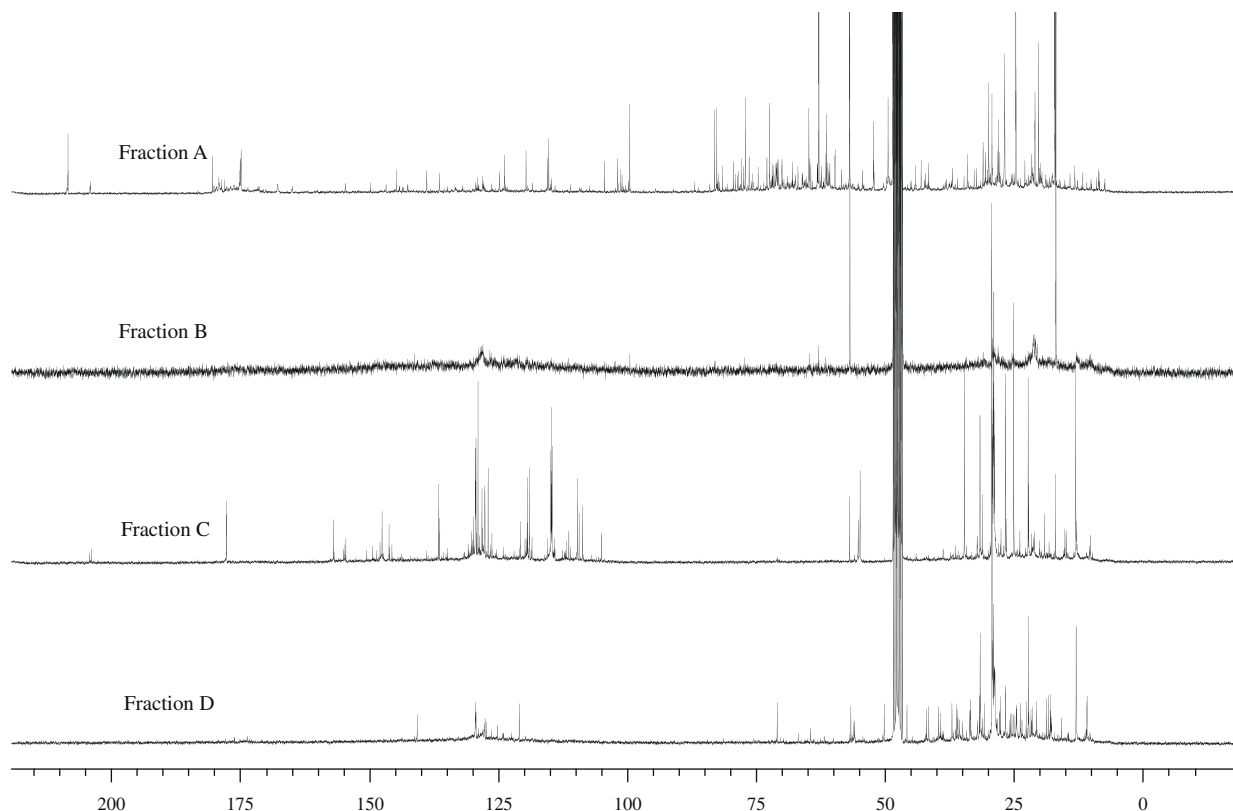


Fig. 4. ^{13}C NMR spectra of fractions A to D.

161), with the $n\text{-C}_7$ the most abundant one (Table 5). Of the basic components, substituted pyrroles, pyrazoles, pyrazines, and pyridines were significant. On the acidic side (Table 6), butyric acid (m/z 87) was a major component, followed by acetic acid (m/z 59), and $n\text{-C}_3$ (m/z 73), $\text{C}_{5:2}$ (m/z 97), $\text{C}_{6:2}$ (m/z 111), $\text{C}_{7:2}$ (m/z 125), $n\text{-C}_7$ (m/z 129) and $n\text{-C}_{11}$ (m/z 185) fatty acids.

Weakly acidic fraction B was dominated by unsaturated fatty acids $\text{C}_{18:1}$ (m/z 281) and $\text{C}_{18:2}$ (m/z 279) (Table 6). In addition, the presence of a series of n -fatty acids, ranging from $n\text{-C}_{14}$ to $n\text{-C}_{28}$, was also observed. Other minor components of this fraction were substituted pyrazole, pyrazines and a nitrile as well as small amounts of diols (Table 6). Of special interest was the detection of dihydro- β -sitosterol (m/z 417). Small amounts of sterols were also observed in Fractions A, C and D.

The basic fraction C (Table 6) contained mainly substituted nitriles, pyrroles, pyrazoles, pyridines and pyrazines. These results are in agreement with the ^1H and ^{13}C NMR analysis data (Figs. 3 and 4), which showed peaks in the region of 6.5 to 7.5 ppm and 106 to 150 ppm, respectively due to the presence of aromatic and N-heterocyclic protons and carbons. In addition, very small amounts of $n\text{-C}_{12}$ (m/z 199) to $n\text{-C}_{18}$ fatty acids (m/z 283), $\text{C}_{6:2}$ fatty acid (m/z 111), $n\text{-C}_4$ (m/z 91) and $n\text{-C}_8$ (m/z 147) to $n\text{-C}_{10}$ diols (m/z 175) and phenols (phenol, guaiacol and alkyl substituted guaiacol) were also detected in this fraction (Table 6).

The neutral fraction D was relatively rich in saturated and unsaturated fatty acids (Table 6). Saturated fatty acids ranged from $n\text{-C}_{14}$ (m/z 227) to $n\text{-C}_{28}$ (m/z 423). Unsaturated fatty acids detected in this fraction were $\text{C}_{5:2}$ (m/z 97), $\text{C}_{6:2}$ (m/z 111), $\text{C}_{18:1}$ (m/z 281), $\text{C}_{18:2}$ (m/z 279) and $\text{C}_{19:1}$ (m/z 295) (Table 5). Other components of this fraction were N-containing compounds such as amides, and alkyl substituted N-heterocyclics such as pyrroles, pyrazoline, pyrazoles, pyrazines and pyridines (Table 6). Fraction D also contained a number of phenols derived from lignin such as coniferyl

alcohol (m/z 181) and sinapyl aldehyde (m/z 209), and the sterol ethyl cholesterol (m/z 397) (Table 6).

Our previous studies of the biooil by Pyrolysis Field Ionization and Pyrolysis Field Desorption mass spectroscopy (Schnitzer et al., 2008) showed the presence of saturated and unsaturated hydrocarbons in biooil (alkanes and alkenes). These compounds

Table 5

Tentative peak identification of the fatty acids in the four separated biooil fractions analyzed by electrospray ionization mass spectrometry in negative mode.

m/z ($m = [\text{M}-\text{H}]^-$)	% -Base Peak Intensity				Compound
	A	B	C	D	
59	7.1	–	–	–	Acetic acid
73	14.3	–	–	–	$n\text{-C}_3$ fatty acid
87	55.5	8.5	–	–	$n\text{-C}_4$ fatty acid
97	28.2	–	–	14.8	$\text{C}_{5:2}$ fatty acid
111	27.1	–	14.7	9.6	$\text{C}_{6:2}$ fatty acid
125	26.8	–	–	–	$\text{C}_{7:2}$ fatty acid
129	16.1	–	–	–	$n\text{-C}_7$ fatty acid
185	9.4	–	–	–	$n\text{-C}_{11}$ fatty acid
199	–	–	11.9	–	$n\text{-C}_{12}$ fatty acid
213	–	–	10.0	–	$n\text{-C}_{13}$ fatty acid
227	–	6.5	12.5	10.2	$n\text{-C}_{14}$ fatty acid
241	–	15.7	–	12.0	$n\text{-C}_{15}$ fatty acid
255	–	61.4	17.3	27.0	$n\text{-C}_{16}$ fatty acid
269	–	7.7	–	3.4	$n\text{-C}_{17}$ fatty acid
279	–	78.4	6.4	24.0	$\text{C}_{18:2}$ fatty acid
281	–	10.0	11.7	36.9	$\text{C}_{18:1}$ fatty acid
283	–	20.4	5.9	8.7	$n\text{-C}_{18}$ fatty acid
295	–	5.5	–	4.2	$\text{C}_{19:1}$ fatty acid
297	–	9.6	–	3.0	$n\text{-C}_{19}$ fatty acid
311	–	3.1	–	1.4	$n\text{-C}_{20}$ fatty acid
339	–	1.6	–	–	$n\text{-C}_{22}$ fatty acid
367	–	2.4	–	0.6	$n\text{-C}_{24}$ fatty acid
381	–	–	–	0.4	$n\text{-C}_{25}$ fatty acid
423	–	0.4	–	0.4	$n\text{-C}_{28}$ fatty acid

Table 6

Classes of compounds and tentative peak identifications of the four separated biooil fractions analyzed by electrospray ionization mass spectrometry in positive mode.

<i>m/z</i> (<i>m</i> = [M+H] ⁺)	%–Base peak intensity				Compounds
	A	B	C	D	
<i>(a) Aliphatic N-compounds</i>					
70	19.5	–	100	58.2	Methylpropyl nitrile
74	–	–	–	34.0	Propylamide
84	–	–	14.2	9.4	Dimethyl propyl nitrile
<i>(b) Aromatic N-heterocyclics</i>					
71	–	–	28.9	14.8	Pyrazoline
81	–	–	–	5.8	Pyrazine
83	–	33.1	4.2	24.7	Methyl pyrazole
95	42.4	–	17.9	–	Methyl pyrazine
96	29.9	–	–	–	Dimethyl pyrrole
97	46.1	–	21.7	–	2,4-Dimethyl imidazole
98	34.3	–	4.0	–	Dione pyrrole
99	–	6.2	14.0	–	Dimethyl dihydro pyrazole
100	15.3	–	–	–	Methyl piperidine
109	15.9	–	–	–	Methyl pyrazine
111	42.4	–	22.8	5.2	1,3,5-Trimethyl pyrazole
125	21.4	–	37.9	8.1	2-Methoxy-3-methyl pyrazine
135	10.0	12.3	20.0	–	Methyl dihydro cyclopentapyrazine
136	12.5	–	–	–	3-Butyl pyridine
137	–	6.1	45.0	8.1	Tetramethyl pyrazine
138	–	–	–	15.8	Diethyl methyl pyrrole
139	18.4	–	71.96	10.6	Diethyl-methyl pyrazole
140	–	–	–	6.0	N-methyl-2-formyl-5-hydroxymethylpyrrole
151	63.7	9.7	61.3	8.0	Diethyl Methyl pyrazine
153	13.2	–	100	–	Triethyl pyrazole
165	–	–	51.6	5.8	Diethyl dimethyl pyrazine
167	–	–	87.8	6.1	Triethyl methyl pyrazole
179	–	–	53.5	6.8	Triethyl methyl pyrazine
181	9.9	4.2	89.4	–	Tetraethyl pyrazole
189	–	–	15.8	–	Diethenyl diethyl pyrazole
193	8.0	–	40.0	9.0	Tetraethyl pyrazine
195	–	–	59.1	8.3	Propyl triethyl pyrazole
207	–	–	7.1	6.2	Triethyl Propyl Pyrazine
209	–	–	15.3	–	Diethyl Dipropyl Pyrazine
210	–	–	–	8.8	Methyl-phenyl-triazolo pyridine
263	–	–	–	4.9	Hexamethyl phenanthrene
<i>(c) Phenolics and benzenes</i>					
109	15.9	–	–	–	Methyl phenol
121	27.2	11.8	–	–	Vinyl phenol
123	15.2	–	35.2	–	Dimethyl phenol
139	18.4	–	72.0	–	Methyl guaiacol
153	13.2	–	–	–	Vanillin
181	9.9	4.2	89.4	–	Coniferyl alcohol
195	8.3	–	–	–	Ferulic acid
209	–	–	15.2	–	Sinapyl aldehyde
359	–	7.5	–	4.8	Eicosyl benzene
443	–	0.3	–	0.3	Hexacosyl benzene
471	–	–	–	0.2	Octacosyl benzene
<i>(d) Diols and sterols</i>					
91			7.9		<i>n</i> -C ₄ diol
133	30.7				<i>n</i> -C ₇ diol
147	8.5				<i>n</i> -C ₈ diol
161	23.0		27.7		<i>n</i> -C ₉ diol
175			22.9		<i>n</i> -C ₁₀ diol
217				5.6	<i>n</i> -C ₁₃ diol
301		3.9			<i>n</i> -C ₁₉ diol
385		1.5			<i>n</i> -C ₂₅ diol
397				51.0	Ethyl cholestene
417		2.0			Dihydro- β -sitosterol
441		0.3			<i>n</i> -C ₂₉ diol
483		0.5			<i>n</i> -C ₃₂ diol

could not be detected by ESI-MS because they do not ionize under the experimental conditions employed. However, evidence of their presence in all four fractions was obtained from the FTIR analysis (Fig. 2), which showed large and sharp bands at between 2971 and 2927 and 2870 and 2854 cm^{−1} due to aliphatic –CH₃ and –CH₂ groups, and the peaks between 1463 and 1455 and 1380 and 1398 cm^{−1} due to the bending vibrations of –CH₂ and –CH₃ groups. Further evidence for the presence of hydrocarbons was also ob-

tained from the ¹³C MAS NMR spectra of all four fractions (Fig. 4), which showed peaks in the region of 0 to 40 ppm due to the paraffinic carbons.

The use of ESI-MS for the analysis of four biooil fractions support and complement the elemental analysis and chemical characterization by FTIR, ¹H and ¹³C NMR, and allowed to identify individual *m/z* signals corresponding to specific molecules within five classes of compounds. For example, the presence of N-hetero-

cyclics in Fraction A and C in relatively large amounts is in harmony with the elemental analyses, which showed the highest N/C ratio for these fractions (Table 3). The latter is further supported by the results obtained from ^1H and ^{13}C NMR. Consistent with the elemental analyses, FTIR and NMR data, a separation of groups of components can be observed from the ESI-MS results for the four different fractions presented in Tables 5 and 6. Further research on the separation of specific chemical compounds in the biooil is underway.

4. Conclusions

An acid–base separation method helped simplify the complexity of more than 500 chemical structures found in biooil by grouping them into four fractions of components with enhanced chemical homogeneity. Five classes of compounds were identified by ESI-MS, namely, fatty acids, aliphatic N-compounds, aromatic N-heterocyclics, phenolic and benzenes, diols and sterols. The strongly acidic Fraction A was rich in phenolic compounds and low molecular weight ($<C_{12}$) saturated fatty acids, whereas the weakly acidic Fraction B mainly contained $>C_{12}$ fatty acids and sterols. The basic Fraction C was primarily composed of N-heterocyclics and a few phenols, although relative small amounts of the heterocyclic-N were also found in the other three fractions. The neutral Fraction D was rich in unsaturated fatty acids with one or two double bonds, long-chain saturated fatty acids ($>C_{20}$), aldehydes and alcohols. The presence of hydrocarbons in the fractions was confirmed by FTIR and NMR analyses.

Acknowledgements

The authors thank Peter Fransham, CEO of Advanced Bio-Refinery Inc., for providing the biooil samples for this experiment. The contribution of Erik Tolonen to the experimental work is highly acknowledged. The authors also gratefully acknowledge the technical assistance of John Nikiforuk and Glen Facey for recording the NMR spectra, Graeme Spiers for phosphorous analysis, Monica Molina for the elemental analyses of carbon, nitrogen and sulfur, and Connie Sauder for typing this manuscript. Funding for this research was provided by the Climate Change Technology and

Innovation Program of Natural Resources Canada and Agriculture and Agri-Food Canada.

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